JDBC

1) Java program to display IP address of a given host name using InetAddress

```
import java.net.*;

public class InetAddressExample {
   public static void main(String[] args) {
      try {
         String host = "www.google.com";

         InetAddress address = InetAddress.getByName(host);

         System.out.println("Host Name: " + address.getHostName());
         System.out.println("IP Address: " + address.getHostAddress());
      }
      catch (Exception e) {
         System.out.println("Error: " + e);
      }
   }
}
```

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2) Java program to display components of a URL

```
import java.net.*;

public class URLExample {
   public static void main(String[] args) {
      try {
         URL url = new URL("https://www.example.com/index.html");

         System.out.println("Protocol: " + url.getProtocol());
         System.out.println("Host: " + url.getHost());
         System.out.println("Port: " + url.getPort());
         System.out.println("File: " + url.getFile());
         System.out.println("Path: " + url.getPath());
      }
      catch (Exception e) {
         System.out.println(e);
      }
   }
}
```

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3) One-way communication using Socket and ServerSocketServer

```
import java.net.*;
import java.io.*;
```

```
public class Server {
   public static void main(String[] args) {
      try {
         ServerSocket server = new ServerSocket(5000);
         System.out.println("Server waiting...");

         Socket socket = server.accept();

         DataInputStream dis = new DataInputStream(socket.getInputStream());

         String msg = dis.readUTF();

         System.out.println("Message from Client: " + msg);

         server.close();
      }
      catch (Exception e) {
         System.out.println(e);
      }
   }
}
```

Client

```
import java.net.*;
import java.io.*;

public class Client {
   public static void main(String[] args) {
      try {
         Socket socket = new Socket("localhost",5000);

         DataOutputStream dos = new DataOutputStream(socket.getOutputStream());

         dos.writeUTF("Hello Server");

         dos.flush();
         socket.close();
      }
      catch (Exception e) {
         System.out.println(e);
      }
   }
}
```

---

4) Multi-client server using sockets

```
import java.net.*;
import java.io.*;

class ClientHandler extends Thread {

   Socket socket;

   ClientHandler(Socket socket) {
      this.socket = socket;
   }

   public void run() {
      try {
         DataInputStream dis = new DataInputStream(socket.getInputStream());

         String msg = dis.readUTF();

         System.out.println("Client says: " + msg);

         socket.close();
      }
      catch (Exception e) {
         System.out.println(e);
      }
   }
}

public class MultiServer {

   public static void main(String[] args) throws Exception {

      ServerSocket server = new ServerSocket(5000);

      System.out.println("Server started...");

      while(true) {
         Socket socket = server.accept();

         ClientHandler ch = new ClientHandler(socket);

         ch.start();
      }
   }
}
```

---

5) Two-way chat application Server

```
import java.net.*;
import java.io.*;

public class ChatServer {
   public static void main(String[] args) throws Exception {

      ServerSocket server = new ServerSocket(5000);
      Socket socket = server.accept();

      DataInputStream dis = new DataInputStream(socket.getInputStream());
      DataOutputStream dos = new DataOutputStream(socket.getOutputStream());
      BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

      String msg1="", msg2="";

      while(!msg1.equals("exit")) {

         msg1 = dis.readUTF();
         System.out.println("Client: " + msg1);

         msg2 = br.readLine();
         dos.writeUTF(msg2);
      }

      socket.close();
      server.close();
   }
}
```

Client

```
import java.net.*;
import java.io.*;

public class ChatClient {

   public static void main(String[] args) throws Exception {

      Socket socket = new Socket("localhost",5000);

      DataInputStream dis = new DataInputStream(socket.getInputStream());
      DataOutputStream dos = new DataOutputStream(socket.getOutputStream());
      BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

      String msg1="", msg2="";

      while(!msg2.equals("exit")) {
```

```
            msg2 = br.readLine();
            dos.writeUTF(msg2);

            msg1 = dis.readUTF();
            System.out.println("Server: " + msg1);
        }

        socket.close();
    }
}
```

---

6) JDBC program to display all records

```
import java.sql.*;

public class DisplayRecords {

    public static void main(String[] args) {

        try {
            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/testdb","root","");

            Statement stmt = con.createStatement();

            ResultSet rs = stmt.executeQuery("SELECT * FROM student");

            while(rs.next()) {

                System.out.println(rs.getInt(1) + " " +
                                   rs.getString(2) + " " +
                                   rs.getInt(3));
            }

            con.close();
        }
        catch(Exception e) {
            System.out.println(e);
        }
    }
}
```

---

7) Insert, Update, Delete using Statement

```
import java.sql.*;

public class CRUDStatement {

   public static void main(String[] args) {

      try {
         Connection con = DriverManager.getConnection(
               "jdbc:mysql://localhost:3306/testdb","root","");

         Statement stmt = con.createStatement();

         stmt.executeUpdate("INSERT INTO student VALUES(1,'Rahul',20)");

         stmt.executeUpdate("UPDATE student SET age=21 WHERE id=1");

         stmt.executeUpdate("DELETE FROM student WHERE id=1");

         System.out.println("Operations performed successfully");

         con.close();
      }
      catch(Exception e) {
         System.out.println(e);
      }
   }
}
```

---

8) PreparedStatement insert & retrieve

```
import java.sql.*;

public class PreparedExample {

   public static void main(String[] args) {

      try {
         Connection con = DriverManager.getConnection(
               "jdbc:mysql://localhost:3306/testdb","root","");

         PreparedStatement ps = con.prepareStatement(
               "INSERT INTO student VALUES(?,?,?)");

         ps.setInt(1,2);
         ps.setString(2,"Amit");
         ps.setInt(3,22);
```

```
        ps.executeUpdate();

        Statement stmt = con.createStatement();

        ResultSet rs = stmt.executeQuery("SELECT * FROM student");

        while(rs.next()) {
            System.out.println(rs.getInt(1)+" "+rs.getString(2));
        }

        con.close();
    }
    catch(Exception e) {
        System.out.println(e);
    }
  }
}
```

---

9) JDBC CRUD + CallableStatement

```
import java.sql.*;

public class CallableExample {

  public static void main(String[] args) {

    try {
        Connection con = DriverManager.getConnection(
              "jdbc:mysql://localhost:3306/testdb","root","");

        PreparedStatement ps = con.prepareStatement(
              "INSERT INTO student VALUES(?,?,?)");

        ps.setInt(1,3);
        ps.setString(2,"Neha");
        ps.setInt(3,21);

        ps.executeUpdate();

        CallableStatement cs = con.prepareCall("{call getStudents()}");

        ResultSet rs = cs.executeQuery();

        while(rs.next()) {
            System.out.println(rs.getInt(1)+" "+rs.getString(2));
        }
```

```
        con.close();
    }
    catch(Exception e) {
        System.out.println(e);
    }
  }
}
```

---

15) ResultSet navigation

```
import java.sql.*;

public class ResultSetNavigation {

  public static void main(String[] args) {

    try {
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/testdb","root","");

        Statement stmt = con.createStatement(
            ResultSet.TYPE_SCROLL_INSENSITIVE,
            ResultSet.CONCUR_READ_ONLY);

        ResultSet rs = stmt.executeQuery("SELECT * FROM student");

        rs.first();
        System.out.println("First: " + rs.getString(2));

        rs.next();
        System.out.println("Next: " + rs.getString(2));

        rs.previous();
        System.out.println("Previous: " + rs.getString(2));

        con.close();
    }
    catch(Exception e) {
        System.out.println(e);
    }
  }
}
```

---

16) Insert using PreparedStatement (Secure)

```
import java.sql.*;

public class InsertPrepared {

   public static void main(String[] args) {

      try {
         Connection con = DriverManager.getConnection(
               "jdbc:mysql://localhost:3306/testdb","root","");

         PreparedStatement ps = con.prepareStatement(
               "INSERT INTO student VALUES(?,?,?)");

         ps.setInt(1,5);
         ps.setString(2,"Kiran");
         ps.setInt(3,23);

         ps.executeUpdate();

         System.out.println("Record Inserted");

         con.close();
      }
      catch(Exception e) {
         System.out.println(e);
      }
   }
}
```

---

17) Retrieve using PreparedStatement

```
import java.sql.*;
import java.util.Scanner;

public class RetrievePrepared {

   public static void main(String[] args) {

      try {

         Scanner sc = new Scanner(System.in);

         System.out.print("Enter ID: ");
         int id = sc.nextInt();

         Connection con = DriverManager.getConnection(
```

```
            "jdbc:mysql://localhost:3306/testdb","root","");

        PreparedStatement ps = con.prepareStatement(
            "SELECT * FROM student WHERE id=?");

        ps.setInt(1,id);

        ResultSet rs = ps.executeQuery();

        while(rs.next()) {
          System.out.println(rs.getInt(1)+" "+rs.getString(2));
        }

        con.close();
      }
      catch(Exception e) {
        System.out.println(e);
      }
    }
}
```

---

18) Types of JDBC Drivers (demonstration)

```
public class JDBCDrivers {

    public static void main(String[] args) {

      System.out.println("Types of JDBC Drivers");

      System.out.println("1. Type 1 – JDBC ODBC Bridge Driver");

      System.out.println("2. Type 2 – Native API Driver");

      System.out.println("3. Type 3 – Network Protocol Driver");

      System.out.println("4. Type 4 – Thin Driver");
    }
}
```

---

19) Execute stored procedure using CallableStatement

```
import java.sql.*;

public class StoredProcedureExample {

    public static void main(String[] args) {
```

```
        try {

            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/testdb","root","");

            CallableStatement cs = con.prepareCall("{call getStudents()}");

            ResultSet rs = cs.executeQuery();

            while(rs.next()) {

                System.out.println(rs.getInt(1)+" "+rs.getString(2));
            }

            con.close();
        }
        catch(Exception e) {
            System.out.println(e);
        }
    }
}
```